```
%% Question 1(b) - CORDIC Performance Analysis
clear; clc; close all;

% Number of CORDIC rotations
N = [10 12 14 16 18 20];

%% Vitis HLS synthesis results

% Resource utilization
LUT  = [2205 2205 2205 2205 2205 2205];
FF   = [1361 1361 1361 1361 1362 1362];
DSP  = [10 10 10 10 10 10];
BRAM = [0 0 0 0 0 0];

% Latency and initiation interval (clock cycles)
latency = [144 170 196 222 248 274];
II = [145 171 197 223 249 275];

% Target clock period = 10 ns
clockPeriod = 10e-9;

% Throughput in results per second
throughput = 1 ./ (II * clockPeriod);

% Convert to thousands of results per second
throughput_k = throughput / 1000;

%% RMSE results from C++ testbench

RMSE_R = [
    0.000001239063636
    0.000000064680641
    0.000000129140460
    0.000000129140460
    0.000000129140460
    0.000000129140460
];

RMSE_Theta = [
    0.001373887993395
    0.000274921942037
    0.000080957448517
    0.000016664818759
    0.000004742075362
    0.000000691487401
];

%% Plot 1 - Throughput
```

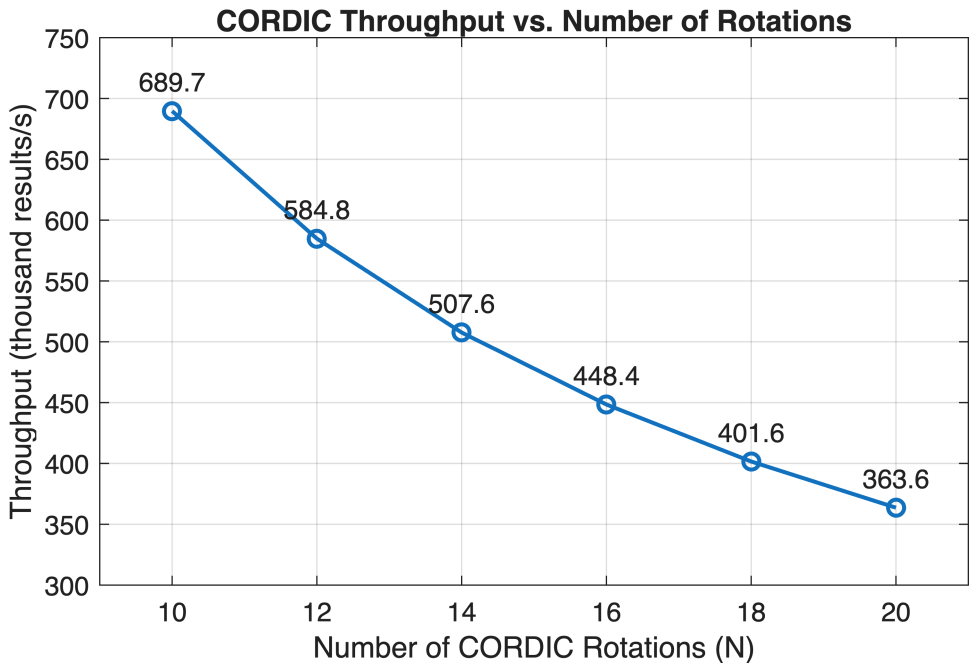
```
figure('Name','CORDIC Throughput');

plot(N, throughput_k, '-o', ...
    'LineWidth', 1.5, 'MarkerSize', 6);

grid on;
xticks(N);
xlim([9 21]);
ylim([300 750]);

xlabel('Number of CORDIC Rotations (N)');
ylabel('Throughput (thousand results/s)');
title('CORDIC Throughput vs. Number of Rotations');

for i = 1:length(N)
    text(N(i), throughput_k(i) + 12, ...
        sprintf('%.1f', throughput_k(i)), ...
        'VerticalAlignment','bottom', ...
        'HorizontalAlignment','center');
end
```




```
%% Plot 2 - FPGA Resource Usage

figure('Name','CORDIC Resource Usage', ...
       'Position',[100 100 1100 600]);

% BRAM = 0 cannot be plotted on a logarithmic axis
BRAM_plot = 0.1 * ones(size(BRAM));

semilogy(N, LUT, '-o', 'LineWidth', 1.5);
```

```
hold on;
semilogy(N, FF, '-s', 'LineWidth', 1.5);
semilogy(N, DSP, '-^', 'LineWidth', 1.5);
semilogy(N, BRAM_plot, '-d', 'LineWidth', 1.5);

grid on;
xticks(N);
xlim([9 21]);
ylim([0.05 10000]);

xlabel('Number of CORDIC Rotations (N)');
ylabel('Resource Count (Log Scale)');
title('FPGA Resource Utilization vs. CORDIC Rotations');

legend('LUT', 'FF', 'DSP', 'BRAM', ...
       'Location','eastoutside');

% Label each data point
for i = 1:length(N)

    % LUT labels
    text(N(i), LUT(i) * 1.2, ...
        sprintf('%d', LUT(i)), ...
        'HorizontalAlignment','center', ...
        'VerticalAlignment','bottom');

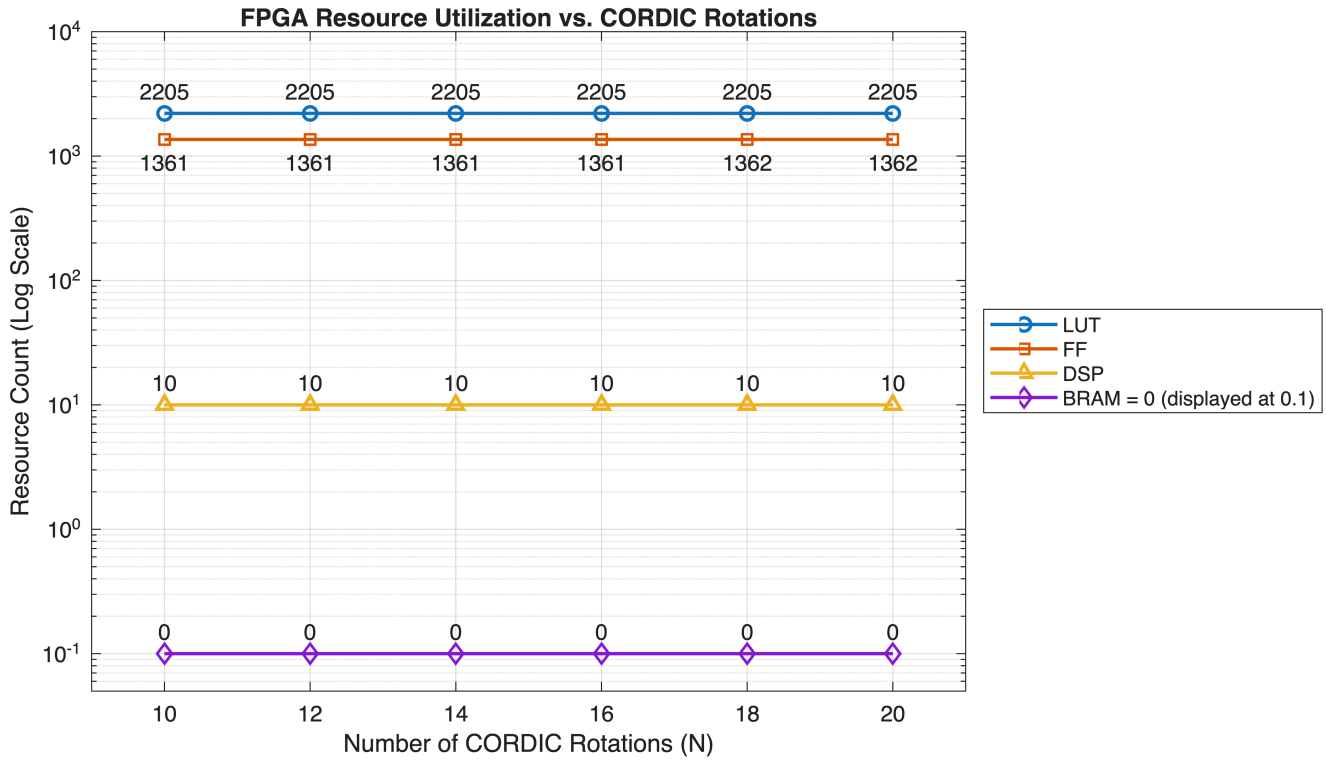
    % FF labels
    text(N(i), FF(i) / 1.2, ...
        sprintf('%d', FF(i)), ...
        'HorizontalAlignment','center', ...
        'VerticalAlignment','top');

    % DSP labels
    text(N(i), DSP(i) * 1.2, ...
        sprintf('%d', DSP(i)), ...
        'HorizontalAlignment','center', ...
        'VerticalAlignment','bottom');

    % BRAM labels (actual value is zero)
    text(N(i), BRAM_plot(i) * 1.2, ...
        sprintf('%d', BRAM(i)), ...
        'HorizontalAlignment','center', ...
        'VerticalAlignment','bottom');

end

hold off;
```




```
%% Plot 3 - RMSE (R and Theta on same plot)

figure('Name','CORDIC RMSE');

semilogy(N, RMSE_R, '-o', ...
    'LineWidth', 1.5, 'MarkerSize', 6);
hold on;

semilogy(N, RMSE_Theta, '-s', ...
    'LineWidth', 1.5, 'MarkerSize', 6);

grid on;
xticks(N);
xlim([9 21]);
ylim([1e-8 1e-2]);

xlabel('Number of CORDIC Rotations (N)');
ylabel('Root Mean Square Error (RMSE)');
title('CORDIC RMSE vs. Number of Rotations');

legend('RMSE(R) - input magnitude units', ...
       'RMSE(Theta) - radians', ...
       'Location','northeast');

% Labels are offset multiplicatively because Y-axis is logarithmic
```

```
for i = 1:length(N)

    text(N(i), RMSE_R(i) / 1.5, ...
        sprintf('%.2e', RMSE_R(i)), ...
        'VerticalAlignment','top', ...
        'HorizontalAlignment','center');

    text(N(i), RMSE_Theta(i) * 1.5, ...
        sprintf('%.2e', RMSE_Theta(i)), ...
        'VerticalAlignment','bottom', ...
        'HorizontalAlignment','center');
end

hold off;
```

